

IEEE 8th VR Workshop on Sonic Interactions for Virtual Environments

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ABSTRACT

Sonic interaction design is defined as the study and exploitation of sound as one of the principal channels conveying information, meaning, and aesthetic/emotional qualities in interactive contexts. This field lies at the intersection of interaction design, and sound and music computing.

In the virtual reality community, the focus of research on topics related to auditory feedback has been rather limited when compared, for example, to the focus placed on visual feedback or even on haptic feedback. However, in such communities as film production or product sound design it is well known that sound is a powerful way to communicate meaning and emotion to a scene or a product.

SIVE 2025 is the 8th of a series of workshops, whose main goal is to increase awareness among the virtual reality community of the importance of sonic elements when designing virtual/augmented/mixed reality (XR hereafter) environments. Participants to the workshop will also discuss how research in other related fields such as film sound theory, product sound design, sound and music computing, game sound design, and accessibility can inform designers of XR environments. Moreover, the workshop will feature state-of-the-art research in the field of sound for XR environments.

Index Terms: H.5.5 [Information interfaces and presentation (e.g., HCI)]: Sound and Music Computing—3D audio, auditory display; H.5.5 [Information interfaces and presentation (e.g., HCI)]: Miscellaneous—eXtended reality, virtual reality, augmented reality;

1 INTRODUCTION

The SIVE workshop, now in its eighth edition, has emerged as a vital platform for exploring the interdisciplinary landscape of sound design, immersive technologies, and human-computer interaction. This year's workshop is particularly significant as it marks the first

edition following the publication of *Sonic Interactions in Virtual Environments (SIVE)* [1], the homonymous book that unifies over a decade of research and practice under a single framework in this domain.

SIVE as a field focuses on integrating auditory stimuli in virtual and augmented environments, emphasizing the role of sound as a primary channel for communication, emotion, and immersion. SIVE investigates how auditory spatiality, interactivity, and multimodal experiences can deepen user engagement and redefine immersion. Unique to this domain is the emphasis on personalization and real-time interaction, which addresses challenges in modeling acoustics, spatial rendering, and auditory perception.

The eighth SIVE workshop will build on the themes established in the book, exploring both theoretical advances and practical applications. By fostering discussions across disciplines such as computer science, acoustics, and psychology, the workshop aims to chart the future trajectory of SIVE as a cornerstone of immersive media technology and design.

2 WORKSHOP PAPERS

The contributions fall into three main categories: Musical Interaction, Cross-modal Perception, and Avatars and Embodiment.

2.1 Musical interaction

Impact of Spatial and Communication Environments on Collaborative Music-Making in Virtual Reality: Insights for Music Education by Michael Oehler, Leonard Bruns, Benedict Saurbier, and Tray Minh Voong, examines the effect of spatial and communication settings on perceived presence, communication quality, and participants' willingness to adopt virtual reality-based collaborative music-making in educational settings. A study conducted with music teachers showed that mediated communication in shared spaces poses significant challenges, however, the participants to the study were positive about the integration of such tools in their educational practice.

In *Developing a 3D interface for sound corpus manipulation in virtual environments*, Matias Vilaplana Stark describes a virtual musical instrument that can be used to interact and reconfigure a collection of sounds in 3D space using different sound descriptors. The results are organized in the 3D space and make the instrument applicable to sound design, music composition, and audiovisual performance.

Virtual Reality Music Training for CI Users: Insights and Design Guidelines by Ali Adjorlu, Emil Sønderskov Hansen, Signe Marie Kromann Kristiansen, and Stefania Serafin, presents a prototype for music training (instrument identification, sound localization

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and melody recognition) in Virtual Reality for Cochlear Implants (CI) users. It relies on an iterative user-centered design approach with a first study on three participants followed by a study on one participant in five sessions, and provides some insights for designing VR applications for CI users.

Room Acoustics affects Immersivity and Musical Behaviour in Virtual Renderings of Real-World Spaces by Marise van Zyl, Matthew Wright, and Takako Fujioka, presents a series of tests in which dry and reverberant auditory interactive feedback was provided to eighteen keyboard players who were virtually immersed in three different performance spaces using a VR headset along with headphones. Results revealed that the presence of room acoustics enhanced participants' sense of immersion and affected their playing-speed and key-press intensity, and pointed to the significance of acoustic reverberation on performance concerning tempo and applied force to the keys.

2.2 Cross-modal perception

In *Effect of Real and Virtual Environmental Context on Loudness Scaling* by Jiajie Xiong, Leny Vincelas, Vit Drga, and Ifat Yasin, a small experiment is presented hypothesizing the existence of cross-modal interactions between loudness and virtual as well as augmented room colors. While no significant difference was found between real and virtual settings, conversely furniture significantly affected loudness perception in a blue room; furthermore, significant effects of color and furnishing were observed for medium to loud sounds amplifying sense of increasing loudness.

Auditory Materiality: Exploring Auditory Effects on Material Perception in Virtual Reality by Bocheng Zhong, Seungwoo Je, and Marcel Zaes Sagesser, contributes to the important question if and how interactive auditory feedback helps render material properties in a virtual scene. A small, qualitative pilot study exploring the perceptual effects of scratching over different materials suggests enhanced immersion and a better understanding of surface properties in the presence of auditory feedback.

In *Measuring Motor Planning using an Affordable Virtual Reality Setup for Accessible Action-Perception Studies*, by Roberto Barumerli and co-authors, the spatial perception of looming sounds (sounds moving toward the user) is studied through consumer-grade head-mounted displays (HMDs). An experiment proved the feasibility of measuring user behavior through this affordable equipment, especially using the reaction times measurement.

2.3 Avatars and Embodiment

In *Investigating the Impact of Avatar Action Sounds on the Plausibility Illusion in Virtual Reality*, by Peter Kurucz, the authors investigate the effect of environmental and action sounds on the plausibility illusion aspect of presence. They conducted a between-subject study with 3 groups of 15 participants who were asked to perform a task in a virtual kitchen under three sound conditions. They then ask participants to complete questionnaires on presence and sickness and reply to open questions regarding their experience.

Exploring the Impact of AI-Generated Speech on Avatar Perception and Realism in Virtual Reality Environments, by Anders Bargum, examines how avatar voices affect participants' perceptions of realism, emotional communication, and social presence. Compared to synthetic voices produced by state-of-the-art literature human-produced voices still represent references. Additionally, Speech-To-Speech techniques performed better than Text-To-Speech techniques.

3 CONCLUSIONS

The 8th edition of the SIVE workshop confirms the interdisciplinarity of the field, showing how sonic interactions are important in several areas of VR such as avatar, embodiment, cross-modal perception and spatial audio.

REFERENCES

- [1] M. Geronazzo and S. Serafin, eds. *Sonic Interactions in Virtual Environments*. Human-Computer Interaction Series. Springer International Publishing, Cham, 1 ed., 2023. doi: 10.1007/978-3-031-04021-4